

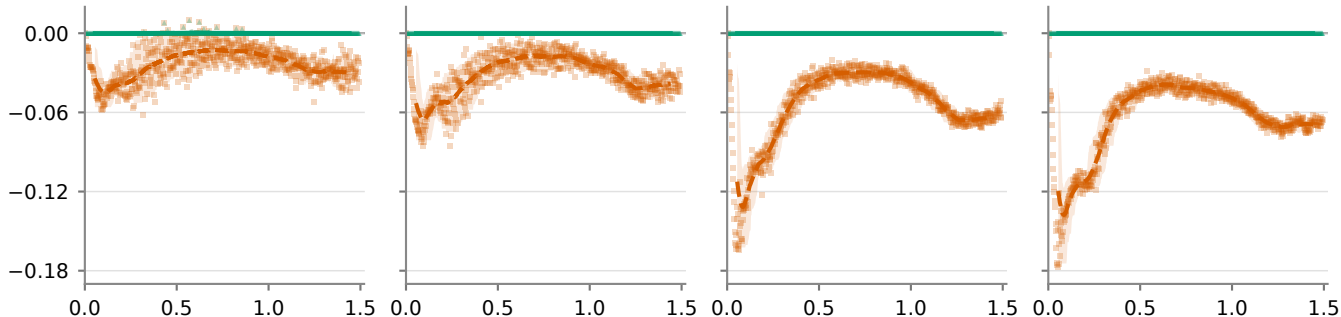
OL1 removes Adam-relative gradient conflict

Before projection After projection
 $\lambda = 0.1$ $\lambda = 0.5$

$\lambda = 0.05$

$\lambda = 1$

Gradient-conflict cosine
(negative = conflict)



Training input tokens (billions)